Personalized Laptop Recommendation Using Decision Tree-Based Classification Algorithm

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***Abstract*—This study developed a supervised learning-based laptop recommendation system using the Decision Tree algorithm combined with *knowledge-based rules* to improve the accuracy and interpretability of prediction results in the TypeName and PriceCategory categories. The data is processed through cleansing, standardization, and feature engineering, then divided in an 80:20 ratio and transformed using *StandardScaler* and *OneHotEncoder*. The test results showed an accuracy of 79%, with superior performance in the premier class, characterized by a 95% recall in the Gaming category, an F1-Score of 0.84 in the Notebook category, and an accuracy of 89% in the Convertible category. Although performance decreases in minority classes due to data imbalances, the integration of knowledge-based rules is able to explain the basis of prediction and provide alternative decisions, making this system effective as a *Decision Support System* in providing accurate and easy-to-understand laptop recommendations.**

Keywords— artificial intelligence, decision tree, supervised learning, laptop recommendation, knowladge-based rules, classification.

I. INTRODUCTION

The development of the portable computing device industry has resulted in a variety of laptop types with increasingly diverse specifications, ranging from basic use needs, productivity needs, to professional needs such as *gaming*, design, and *content creation* (Kotler & Keller, 2016) [30]. This condition makes the process of choosing a laptop no longer simple, because consumers must consider the balance between performance, durability, price, and purpose of use. However, most users do not have the technical ability to interpret device specifications precisely, so purchasing decisions are often made based on brand perceptions or informal recommendations that are not always accurate (Sharma & Kumar, 2020) [24].

In the realm of artificial intelligence, the supervised learning approach has been widely applied to perform hardware feature-based classifications, including laptop type classification and price categories (Jain et al., 2000) [21]. The performance of this learning model is strongly influenced by the quality of processing and *feature engineering* which includes value cleaning, scale transformation, pattern extraction, and categorical attribute conversion (Guyon & Elisseeff, 2003) [8]. In this study, the process is carried out through *a preprocessing script* that includes normalization of RAM and Weight attributes, as well as the extraction of relevant features from ScreenResolution, CPU, GPU, and Memory to be ready for use for model training.

The use of Decision Tree as the main model is based on its interpretable nature, which is able to present predictive results in the form of logical structures that can be explained (Quinlan, 1986) [4]. This is important because models with high interpretability provide transparency in decision-making, especially when used as the basis for product recommendations (Breiman et al., 1984) [6]. In contrast to complex models such as XGBoost or Neural Networks which are often considered *black boxes*, Decision Trees allow users to search for prediction reasons directly (Molnar, 2019) [29]. However, statistical learning-based models alone are not always enough to handle variations in real usage conditions. Therefore, this study adds a component of knowledge-based rules designed based on the results of data exploration and feature correlation. For example, high-capacity RAM tends to appear in gaming laptops or workstations, while lightweight laptops with SSD storage are more often found in the ultrabook category. Such a hybrid approach has long been used in intelligent systems to improve adaptability and decision consistency in cases that are not fully captured by statistical models (Zadeh, 1965 [16]; Burke, 2002 [15]). Furthermore, the formation of the PriceCategory price category uses a *default binning* scheme (Cheap–Medium–Expensive) based on Price_in_euros value provides a stable and market-relevant target structure. This is in line with the concept of consumer segmentation in marketing science, which explains that price ranges are usually directly related to the perception of quality and class of device usage (Kotler & Keller, 2016) [30].

Thus, this research focuses on two main objectives:

1. Build a laptop specification classification model that can identify device categories based on a combination of technical attributes.

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2. Provide a knowledge-based explanation that helps explain the reasoning behind the model's predictions in a clear, structured, and accountable manner.

These two goals are expected to present a laptop recommendation system that is not only accurate, but also easy to understand by end users.

II. Related works

A. Classification on Tabular Data

Classification is the process of mapping a set of features into a label or category using patterns learned from training data (Bishop, 2006) [9]. In the case of laptops, specifications such as CPU, GPU, RAM, screen size, and storage type can be viewed as a representation of features that describe the performance of the device (Sharma & Kumar, 2020) [24]. However, the performance of classification models on tabular data is greatly influenced by feature engineering steps, which are the way features are cleaned, broken down, or encoded before entering the model (Guyon & Elisseeff, 2003) [8]. This is also seen in the processing of the dataset in this study, where columns such as ScreenResolution, Cpu, Gpu, and Memory are elaborated into derived attributes so that the model can read its characteristics clearly.

B. The Role of Feature Engineering in Model Accuracy

Feature engineering not only aims to simplify the data structure, but also improve *the signal-to-noise ratio* in learning (Chandrashekar & Sahin, 2014) [27]. In this dataset, steps such as:

1. convert RAM and Weight to pure numerics.
2. create Touchscreen and IPS indicators from ScreenResolution.
3. and mapping storage types into binary features such as SSDs and HDDs, helping models to capture the relationship between specifications and laptop types.

The use of **StandardScaler** and **OneHotEncoder techniques** through **ColumnTransformer** makes data of various shapes consistent for the model to learn. This is in line with the general guidelines for preprocessing on tabular data (Kuhn & Johnson, 2013) [26].

C. Decision Tree as a Learning Model

Decision Tree is a supervised learning model that expresses the classification process in the form of a decision tree (Quinlan, 1986) [4]. This model is widely used because it is easy for humans to understand and can be interpreted directly (Molnar, 2019) [29]. In addition, Decision Tree does not require data distribution assumptions and is capable of handling both numerical and categorical features on the same model (Witten et al., 2011) [22].

In this study, the use of Decision Tree is very appropriate because the prediction results can be clearly explained when the system is used as a recommendation for laptop choice.

D. Knowledge-Based Rules for Interpretability

Although Decision Tree models have high interpretability, real-world situations often require additional reasoning when the model's predictions are at overlapping boundaries or classes (Burke, 2002) [15]. For this reason, a knowledge-based approach is used as a complement. The principle is similar to the classical expert system that makes decisions based on rules derived from the understanding of the human domain (Zadeh, 1965) [16]. In this study, rules are formulated from correlation patterns in data exploration, for example:

1. The higher the RAM, the stronger the orientation to gaming or content creation.
2. Laptops weighing < 1.5 kg and using SSDs are usually in the ultrabook category**.**
3. The PriceCategory class with the lowest value most often has to do with 8 GB of RAM ≤ and GPU integration.

These rules were formulated from the EDA analysis of the dataset.

E. Price Segmentation and User Class

The division of price categories into Cheap – Medium – Expensive is not only technical, but also related to the perception of value and the market (Kotler & Keller, 2016) [30]. Using the default price splitting, the model can learn about specification patterns that align with the user's needs segment without additional special treatment of the dataset.

III. Methodology

This research methodology is designed so that the classification process of laptop specifications can be carried out systematically, can be replicated, and produces measurable outputs. The main approach consists of four stages: data capture, preprocessing and feature engineering, Decision Tree model training, and application of knowledge-based rules to support predictive interpretability.

A. Dataset

The data used comes from the Laptop Price Dataset available on the Kaggle platform (Kaggle, 2020) [23]. This dataset contains hardware-related attributes such as CPU, GPU, RAM, storage type, screen size, weight, and price in Euro units. Each of these attributes has a different influence on the category of laptop usage (Sharma & Kumar, 2020) [24]. In the early stages, columns that don't contribute to the classification process such as Product and laptop_ID are removed to prevent analysis bias. This is in accordance with data preprocessing guidelines to eliminate irrelevant features (Han, Kamber & Pei, 2012) [10].

B. Preprocessing and Feature Engineering

The preprocessing stage is carried out based on the script that has been developed. Feature engineering is done with the aim of making the data more informative and easier to learn models (Guyon & Elisseeff, 2003) [8].

The steps taken include:

TABLE I
STEPS OF PREPROCESSING AND FEATURE ENGINEERING

Steps	Purpose
Conversion of RAM and Weight to pure numerical	Omit the unit of text for calculation
Touchscreen and IPS Extraction from ScreenResolution	Unveiling the characteristics of the screen
Simplification of CPU Brand and GPU Brand	Reduce sparseness categories to avoid overfitting
Memory mapping to SSD/HDD/Flash/Hybrid features	Capture differences in storage media performance
PriceCategory Formation (Cheap – Medium - Expensive)	Provide clear price classification targets

Furthermore, numeric features are processed using StandardScaler, while categorical features are converted into numerical representations via OneHotEncoder. All transformations are performed in a structured manner using ColumnTransformer, which ensures uniformity of output and minimizes processing errors. This approach is in line with preprocessing practices on tabular data models (Kuhn & Johnson, 2013) [26].

The dataset was then divided into 80% of the training data and 20% of the test data using the stratified split technique to keep the label distribution balanced. All processing results are saved into a CSV file so that they can be replicated without repeating the preprocessing stage.

C. Learning Model: Decision Tree

The main model used is the Decision Tree (CART) due to its high interpretability (Quinlan, 1986) [4] and its ability to handle both numerical and categorical features simultaneously (Witten, Frank & Hall, 2011) [22]. A pattern learning model based on the selection of attributes that provide the best separation between classes uses a measure of Gini Impurity (Breiman et al., 1984) [6].

Advantages of this approach:

1. It is easy to explain the reason for the prediction to the user (Molnar, 2019) [29].
2. It does not require the assumption of data distribution.
3. Suitable for specification-based recommendation systems.

However, the Decision Tree may experience overfitting, so the evaluation results will be tested on the test data to ensure that generalizations are maintained.

D. Knowledge-Based Rules

To improve the clarity of prediction results, the model is equipped with domain-knowledge-based rules, which are formulated from the results of data exploration and feature correlation.

Examples of rules applied:

TABLE II
ATURAN KNOWLEDGE-BASED RULES

Feature Conditions	Recommended Categories
RAM ≥ 16 & GPU Brand = NVIDIA/AMD	Gaming / Content Creation
Weight ≤ 1.5 kg & SSD = 1	Ultrabook / Office
PriceCategory = Cheap & RAM ≤ 8	General / Entry Level Use

This rule acts as:

1. Post-prediction explanation (the user knows the reason for the decision).
2. Tie-breaker when the probability of the model's prediction is low.

This hybrid approach is also applied to modern recommendation systems (Burke, 2002).

IV. RESULT AND DISCUSSION

A. Decision Tree Model Performance

The Decision Tree (CART) model is trained using X_train_processed and tested on X_test_processed, which has gone through the scaling and one-hot encoding stages to ensure the input is in a consistent format. This allows the model to perform the best attribute selection at each node separation process based on the Gini Impurity index (Breiman et al., 1984) [6].

Attributes that consistently appear as the main debiters on top-level nodes include:

1. RAM
2. GPU Brand
3. Storage Type (SSD/HDD)
4. Screen Size (inches)

These findings are in line with the literature that states that these specifications are the most direct indicators of laptop performance and usage class (Sharma & Kumar, 2020) [24].

In other words, the results of the model training not only follow statistical patterns, but also make technical sense, thus strengthening the validity of the model.

B. The Role of Knowledge-Based Rules in Interpretability

Although Decision Tree is interpretable, there are conditions when the boundaries between laptop categories intersect e.g. 14-inch laptops with 8–16 GB of RAM can be categorized as Ultrabook, Office, or General Use depending on the GPU and price. This is where the rules come into play.

Rules such as:

TABLE III
RULES AND THE ROLE OF KNOWLEDGE-BASED RULES IN INTERPRETABILITY

Conditions	System Interpretation
RAM ≥ 16 & GPU Brand = NVIDIA/AMD	Laptops are geared towards Gaming/Content Creation
SSD = 1 & Weight ≤ 1.5 kg	Laptop is geared to Ultrabook / Office Mobility
PriceCategory Cheap & RAM ≤ 8	General Use / Entry Tier

This rule is taken from data correlation patterns and feature exploration. In theory, this approach combines data-driven learning and domain reasoning, following the concept of hybrid intelligent systems (Burke, 2002) [15].

The advantage is that users can see the reason for the prediction explicitly, rather than simply receiving results without explanation, an important aspect of the decision support system.

C. Advantages of the Hybrid Approach

TABLE IV
ADVANTAGES OF THE HYBRID APPROACH

Dimensions	Models Only (Decision Tree)	Model + Knowledge-Based Rules
Interpretability	Tall, but limited to tree structure	Very high, because the rules are easy to explain
Predictive Stability	Good on clear data	Better on ambiguous data
Suitability for recommendations	Enough	Optimal, because there are predictive reasons that accompany the results
Data dependency	Height	Medium, because the rules can cover the lack of data

This hybrid approach is also consistent with the idea that user-acceptable AI should be able to explain the "why" (Ribeiro et al., 2016) [13].

D. System Limitations

While this approach is powerful, there are some limitations:

1. Decision Trees are prone to overfitting if the size of the tree is too deep (Breiman, 2001) [7].
2. Knowledge-based rules require expert validation so that they are not biased or too rigid.
3. If the dataset is extended with a new generation of laptops, the rules may need to be updated, for example the latest series GPUs or LPDDR5 memory configurations.

This limitation is natural and becomes room for improvement in the next stage of development.

E. Practical Implications

This approach is best suited to:

1. Laptop recommendation system in online stores.
2. Device selection assistance for IT consultants.
3. Laptop purchase assistant for non-technical users.

Because this system not only gives results, but also gives reasons that are easy to understand.

V. CONCLUSION

This study shows that the process of classifying laptop specifications can be carried out effectively by combining the supervised Decision Tree model and knowledge-based rules to improve the accuracy of system understanding and interpretability.

The Decision Tree model used is able to learn the pattern of relationships between device specifications and laptop categories without requiring complex data distribution assumptions, resulting in a decision structure that can be traced directly by users.

The preprocessing stage, which includes attribute cleaning, feature engineering such as Touchscreen extraction, IPS, and storage type mapping, as well as StandardScaler and OneHotEncoder, plays an important role in maintaining the quality of the model input.

In addition, the formation of PriceCategory based on default binning helps provide a structured and relevant classification target to market segmentation.

The addition of knowledge-based rules provides significant added value because it allows the system to explicitly explain the reasons behind the predictions, for example through performance indicators such as RAM, GPU brand, and storage type. This hybrid approach is aligned with the principles of intelligent systems that balance learning from data and domain knowledge. Thus, the resulting system not only provides predictive results, but also provides a logical basis that can be understood, making it suitable for use in the context of device recommendation applications, e-commerce platforms, and laptop purchase decision support systems.

Based on the results of the study, the laptop recommendation system developed by integrating Decision tree algorithms and *knowledge-based rules* is proven to be able to provide good classification performance and easy-to-interpret results. The model achieves an accuracy of 79% with superior performance in the premier class, demonstrated by a 95% recall value in the Gaming category, an F1-Score of 0.84 in the Notebook category, and an 89% precision in the Convertible category, reflecting the model's ability to map technical specifications effectively. Despite the limitations of the minority class due to data imbalances, the rule-based approach is able to complement the model with a logical explanation of the predicted outcome and provide an

alternative decision when confidence levels are low. Thus, the proposed system is feasible to be implemented as a *Decision Support System* to assist users in choosing a laptop more accurately, rationally, and transparently.

Future development may include the implementation:

1. ensemble models such as Random Forest or XGBoost to improve accuracy
2. application of SHAP or LIME-based interpretability for more detailed visualization of prediction reasons.

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